Effort, Time, and Cost Estimation for OLMS:

Since **OLMS is a Semi-Detached system**, we will use the **COCOMO Basic Model** for effort, time, and cost estimation.

Step 1: ***Use the COCOMO Basic Model Equations***

For **Semi-Detached** systems, the estimation formulas are:
Effort (Person-Months):

$$Effort = 3.0 \times (KLOC)^{1.12}$$

Time (Months):

$$Time = 2.5 \times (Effort)^{0.35}$$

Cost (in Rupees):

$$Cost = Effort \times Average\ Salary$$

Step 2: ***Convert Function Points (FP) to KLOC (Kilo Lines of Code)***

- To estimate **KLOC**, we assume:
- Adjusted Function Points (AFP) = 129
- **Average FP per KLOC** = **100** (for business applications)

$$KLOC = \frac{AFP}{100} = \frac{129}{100} = 1.29$$

Step 3: ***Calculate Effort (E):***

$$Effort = 3.0 \times (1.29)^{1.12}$$

$$Effort = 3.0 \times 1.33$$

$$Effort = 3.99 \approx 4\ (Person - Months)$$

Step 4: ***Calculate Time (T):***

$$Time = 2.5 \times (4)^{0.35}$$

$$Time = 2.5 \times 1.63$$

$$Time = 4.07 \approx 4.1\ Months$$

Step 5: ***Calculate Cost:***

If the Average Salary per Developer per Month is ₹60,000, then:

$$Cost = 4 \times 60{,}000$$

$$Cost = 2{,}40{,}000$$

This estimation helps in planning the budget, resources, and timeline for OLMS development under the **Agile Model with a Semi-Detached COCOMO approach**.

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